

KIRITI AMARAVADI

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EXPERIENCE

Goldman Sachs

Sep 2025 – Present

Software Engineer, Financial Engineering

Dallas, TX

- Lead engineer on a Contract Lifecycle Management platform, a microservices system processing vendor contracts through LLM-powered digitization, semantic document search, and automated approval workflows deployed on AWS.
- Built workflow engines using Spring State Machine for multi-stage approval processes; engineered bulk data ingestion pipelines with per-row error isolation and compensating rollback; implemented async reporting infrastructure using CompletableFuture.
- Designed a serverless AI pipeline (Step Functions + Lambda) running Textract OCR, LangChain/LangGraph structured attribute extraction (GPT-4.1-mini), and OpenSearch neural indexing; EventBridge-triggered reconciliation with automated HTML audit reports.
- Built multi-modal document search service: natural language queries via LLM intent extraction to OpenSearch DSL, structured predicate search with dynamic index mapping resolution, and multi-turn conversational RAG using Gemini-2.5-flash with LRU session management.
- Developed React 19 SPA with AI-powered document discovery, LLM-assisted forms, and PDF approval workflows; provisioned full cloud infrastructure as code via AWS CDK (TypeScript). Secured services with OAuth2 JWT and mTLS; monitored via CloudWatch with PagerDuty alerting.

U Control Billing

Nov 2024 – Mar 2025

Software Engineer

Healthcare Technology

- Integrated an ML-based anomaly detection module to flag irregular claim patterns (duplicate submissions, outlier billing codes, payer mismatches) before entering the processing queue; improved throughput by **35%** through parallel claim batching and query optimization.
- Built a pattern recognition layer classifying incoming claims by type, payer, and risk score to route them through differentiated Kafka validation pipelines; developed React.js dashboards with real-time WebSocket feeds, anomaly trend charts, and a configurable rule engine UI.

Democracy Lab

Jul 2024 – Feb 2025

Software Engineer

Civic Technology

- Built an NLP-powered ticket routing engine (TF-IDF + classifier) predicting category and urgency from submission text, automatically assigning tickets to the correct team and reducing manual triage by **60%**.
- Designed event-driven architecture with Kafka for cross-service communication; implemented SLA escalation logic with configurable time-window rules and WebSocket-push alerts; built a React dashboard with live queue views, SLA countdown indicators, and RBAC via Spring Security + OAuth 2.0.
- Reduced release cycle time by **40%** through CI/CD automation and Kubernetes rolling deployments on AWS; achieved **95%** API contract coverage through integration tests.

TECHNICAL SKILLS

Languages: Java 21, Python 3.12, TypeScript, Swift, Kotlin

Backend & Frameworks: Spring Boot, FastAPI, Spring State Machine, Spring Security, Spring Cloud OpenFeign, React 19

AI / LLM: LangChain, LangGraph, GPT-4.1-mini, Gemini-2.5-flash, RAG, Vector Search

AWS: Step Functions, Lambda, ECS Fargate, OpenSearch, Aurora Serverless v2, SQS/DLQ, Textract, CDK

Data & Storage: PostgreSQL, Redis, OpenSearch k-NN, Hibernate / JPA

DevOps & Quality: Docker, GitLab CI/CD, SonarQube, JUnit 5, Mockito, pytest, Vitest, JaCoCo

PROJECTS

AI-Powered Document Intelligence Pipeline | *Python, LangChain, LangGraph, Step Functions, OpenSearch*

- Serverless pipeline orchestrated by Step Functions and Lambda: OCR via Textract, LLM-based structured data extraction (LangChain/LangGraph + GPT-4.1-mini), and OpenSearch neural indexing with k-NN vector embeddings.

- Multi-modal search service: natural language queries processed by an LLM intent analyzer generating OpenSearch DSL; structured predicate search with dynamic index field resolution; multi-turn conversational RAG using Gemini-2.5-flash with LRU session management.

Contract Lifecycle Management Platform | Java 21, Spring Boot, React 19, Aurora Serverless v2, AWS CDK

- Spring State Machine workflow engine enforcing multi-stage approval processes; bulk ingestion pipeline with per-row error isolation and compensating rollback; document repository with content-based deduplication and SQS-driven async OpenSearch indexing.
- React 19 SPA with AI-powered document discovery, LLM-assisted data entry, side-by-side document approval workflows, and analytics dashboards; full cloud infrastructure provisioned as code via AWS CDK across multiple stacks.

EDUCATION

University of North Carolina at Charlotte <i>Master of Science, Computer Science</i>	Jan 2023 – May 2024 <i>Charlotte, NC</i>
Guru Nanak Institutions Technical Campus <i>Bachelor of Technology, Computer Science & Engineering</i>	Aug 2018 – Aug 2022 <i>Hyderabad, India</i>